



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AATHETYA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

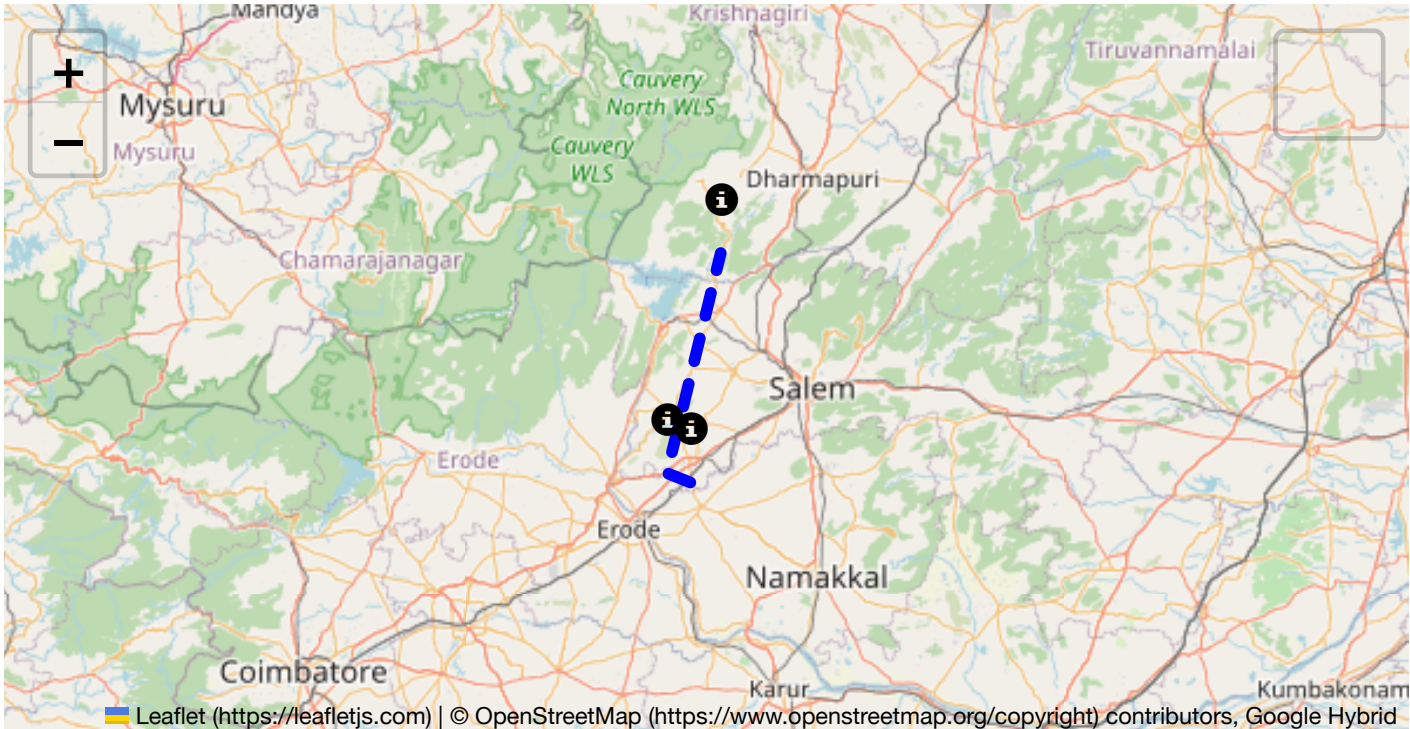
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 78.08 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours

Start: (11.4381, 77.8734)
End: (11.964225, 77.940457)



Welcome to the Journey Risk Management Study

Route Overview

The route covers approximately 78.08 kilometers, beginning at CVQF+23W, Sangagiri, and traveling through FR48+7XQ, Bhavani Main Road in Padaiveedu before reaching the destination at Pennagaram Main Road, Arakasanahalli, Tamil Nadu. The journey is typically around 1.93 hours for heavy vehicles with hazardous materials.

Weather Conditions

Typical Conditions:

- **Summer (March - June):** Hot and dry with temperatures ranging from 30°C to 40°C.
- **Monsoon (July - September):** Heavy rainfall leading to slippery roads and possible flooding.
- **Winter (October - February):** Mild with temperatures between 20°C to 25°C.

Potential Hazards:

- **Monsoon** can cause reduced visibility and waterlogged roads.
- **Summer heat** may lead to tire blowouts and vehicle overheating.

Traffic Patterns

Peak Hours:

- **Morning:** 8:00 AM - 10:00 AM
- **Evening:** 5:00 PM - 8:00 PM

Congestion Areas:

- Bhavani Main Road is prone to congestion, particularly around market areas and towns due to high local traffic and narrow roads.

Road Quality and Infrastructure

- **General Condition:** Mixed quality with sections well-maintained and others with potholes and uneven surfaces.
- **Infrastructure:** Adequate signage, moderate width roads, some areas lacking shoulders.

Alternative Routes for Emergencies

- Consider detouring through NH44 if Padaiveedu and surrounding roads are congested or blocked, although it may increase travel distance.

Local Regulations for Hazardous Materials

- Adhere to local transport regulations including obtaining the necessary permits.
- Transport of hazardous materials is typically restricted during peak traffic hours and in school zones.

Historical Incidents

- **Data** on incidents involving heavy vehicles is sparse, but caution is advised in accident-prone areas like narrow village roads.

Environmental Considerations

- **Sensitive Regions:** Close proximity to agricultural lands; spills could lead to soil and water contamination.
- **Wildlife Crossings:** Be cautious near forested areas for animal crossings.

Communication Coverage

- Coverage is generally reliable along the route, but potential dead zones exist in rural and hilly areas.
- Ensure communication devices are fully charged and emergency contacts are on speed dial.

Emergency Response Times

- **Urban Areas:** 15-30 minutes
- **Rural Areas:** 30-60 minutes, depending on terrain and accessibility.

Risk Assessment Summary

- **Overall Risk Level:** Moderate
- **Key Concerns:** Weather conditions in monsoon, road congestion, variable road quality, and environmental impact in case of spills.
- **Recommendation:** Regularly monitor weather forecasts, maintain vehicle safety checks, and establish clear communication lines.

By adhering to the above guidelines and staying informed of route conditions, drivers can mitigate risks associated with transporting hazardous materials along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45853, 77.85139	30 KM/Hr
7	Turn	Medium	11.45906, 77.85132	30 KM/Hr
8	Turn	High	11.46307, 77.84941	15 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45636, 77.81678	15 KM/Hr
11	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
12	Turn	Medium	11.49441, 77.86690	30 KM/Hr
13	Turn	Medium	11.49971, 77.86468	30 KM/Hr
14	Turn	Medium	11.50885, 77.86037	30 KM/Hr
15	Turn	Medium	11.52519, 77.85535	30 KM/Hr
16	Turn	Medium	11.52744, 77.85582	30 KM/Hr
17	Turn	Medium	11.55499, 77.85656	30 KM/Hr
18	Turn	High	11.57893, 77.85430	15 KM/Hr
19	Turn	Medium	11.57919, 77.85585	30 KM/Hr
20	Turn	Medium	11.58037, 77.85749	30 KM/Hr
21	Turn	Medium	11.60978, 77.84165	30 KM/Hr
22	Turn	Medium	11.61795, 77.84020	30 KM/Hr
23	Turn	Medium	11.63085, 77.84072	30 KM/Hr
24	Turn	Medium	11.64073, 77.84627	30 KM/Hr
25	Turn	Medium	11.64130, 77.84816	30 KM/Hr
26	Turn	Medium	11.64547, 77.85129	30 KM/Hr
27	Turn	Medium	11.65486, 77.85335	30 KM/Hr
28	Turn	Medium	11.66653, 77.85622	30 KM/Hr
29	Turn	High	11.67310, 77.86031	15 KM/Hr
30	Turn	Medium	11.67357, 77.86193	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	11.68196, 77.86690	30 KM/Hr
32	Turn	High	11.68492, 77.86898	15 KM/Hr
33	Turn	High	11.68463, 77.87262	15 KM/Hr
34	Turn	High	11.69455, 77.87583	15 KM/Hr
35	Turn	High	11.69558, 77.87439	15 KM/Hr
36	Turn	High	11.76019, 77.89047	15 KM/Hr
37	Turn	High	11.75964, 77.89200	15 KM/Hr
38	Blind Spot	Blind Spot	11.83264, 77.94408	10 KM/Hr
39	Turn	Medium	11.84317, 77.93283	30 KM/Hr
40	Turn	Medium	11.84593, 77.92925	30 KM/Hr
41	Turn	Medium	11.85102, 77.92587	30 KM/Hr
42	Turn	Medium	11.88158, 77.93124	30 KM/Hr
43	Turn	Medium	11.88495, 77.93258	30 KM/Hr
44	Turn	Medium	11.88951, 77.93278	30 KM/Hr
45	Turn	High	11.88979, 77.93330	15 KM/Hr
46	Turn	Medium	11.89093, 77.93364	30 KM/Hr
47	Turn	Medium	11.89304, 77.93241	30 KM/Hr
48	Turn	Medium	11.89367, 77.93241	30 KM/Hr
49	Turn	Medium	11.89447, 77.93194	30 KM/Hr
50	Turn	Medium	11.89594, 77.93195	30 KM/Hr
51	Turn	High	11.89818, 77.93069	15 KM/Hr
52	Turn	Medium	11.89893, 77.93090	30 KM/Hr
53	Turn	Medium	11.89967, 77.93212	30 KM/Hr
54	Turn	Medium	11.90023, 77.93247	30 KM/Hr
55	Turn	Medium	11.90083, 77.93245	30 KM/Hr
56	Turn	Medium	11.90434, 77.93007	30 KM/Hr
57	Turn	High	11.90535, 77.92665	15 KM/Hr
58	Turn	Medium	11.90979, 77.92570	30 KM/Hr
59	Turn	High	11.91187, 77.92304	15 KM/Hr
60	Turn	High	11.91332, 77.92336	15 KM/Hr
61	Turn	High	11.91408, 77.92286	15 KM/Hr
62	Turn	Medium	11.91664, 77.92269	30 KM/Hr
63	Blind Spot	Blind Spot	11.92989, 77.92663	10 KM/Hr
64	Blind Spot	Blind Spot	11.93022, 77.92591	10 KM/Hr
65	Turn	Medium	11.94153, 77.92988	30 KM/Hr
66	Turn	Medium	11.94365, 77.92965	30 KM/Hr
67	Turn	Medium	11.94447, 77.93046	30 KM/Hr
68	Turn	Medium	11.94615, 77.93045	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
69	Turn	Medium	11.94962, 77.93394	30 KM/Hr
70	Turn	High	11.95109, 77.93458	15 KM/Hr
71	Turn	Medium	11.95167, 77.93416	30 KM/Hr
72	Turn	Medium	11.95390, 77.93454	30 KM/Hr
73	Turn	High	11.95462, 77.93515	15 KM/Hr
74	Turn	High	11.95442, 77.93544	15 KM/Hr
75	Blind Spot	Blind Spot	11.95363, 77.93517	10 KM/Hr
76	Turn	High	11.95642, 77.93570	15 KM/Hr
77	Turn	High	11.96346, 77.93881	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
9	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
10	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
13	hospital	Rishan Hospital	11.879358, 77.9286133	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49441, 77.86690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49971, 77.86468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50885, 77.86037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52519, 77.85535



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52744, 77.85582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55499, 77.85656



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57919, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58037, 77.85749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60978, 77.84165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63085, 77.84072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64073, 77.84627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64130, 77.84816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64547, 77.85129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65486, 77.85335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66653, 77.85622



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67310, 77.86031



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67357, 77.86193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68492, 77.86898



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69455, 77.87583



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69558, 77.87439



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75964, 77.89200



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.83264, 77.94408



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84317, 77.93283



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84593, 77.92925



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.85102, 77.92587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88158, 77.93124



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88495, 77.93258



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88951, 77.93278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.88979, 77.93330



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89093, 77.93364



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89304, 77.93241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89367, 77.93241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89447, 77.93194



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89594, 77.93195



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.89818, 77.93069



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89893, 77.93090



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89967, 77.93212



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90023, 77.93247



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90083, 77.93245



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90434, 77.93007



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.90535, 77.92665



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90979, 77.92570



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91187, 77.92304



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91332, 77.92336



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91408, 77.92286



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91664, 77.92269



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.92989, 77.92663



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.93022, 77.92591



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94153, 77.92988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94365, 77.92965



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94447, 77.93046



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94615, 77.93045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94962, 77.93394



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95109, 77.93458



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95167, 77.93416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95390, 77.93454



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95462, 77.93515



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95442, 77.93544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.95363, 77.93517



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95642, 77.93570



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96346, 77.93881
